



GOSFORD HIGH SCHOOL

September 2019

Preliminary Mathematics Extension 1

Assessment Task 3

General Instructions

- Reading time – 5 minutes
- Working time – 1 hour 30 minutes
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- All necessary working should be shown in Questions 9 - 25

Total Marks – 55

Section I - 8 marks

- Attempt Questions 1 - 8
- Allow about 12 minutes for this section

Section II - 47 marks

- Attempt Questions 9 - 15
- Allow about 1 hour 18 minutes for this section

STUDENT NUMBER/NAME:

Section I

8 Marks

Attempt Questions 1-8

Allow about 12 minutes for this section.

Select the alternative A, B, C or D that best answers the question and indicate your choice with a cross (X) in the appropriate space on the grid below.

	A	B	C	D
1				
2				
3				
4				
5				
6				
7				
8				

Section I

8 Marks

Attempt Questions 1-8

Allow about 12 minutes for this section.

Use the multiple-choice answer sheet for questions 1-8

-
1. What is the Cartesian equation of the line with parametric equations $x = t + 2$ and $y = 2t + 1$? 1
- (A) $2x + y + 3 = 0$
(B) $2x - y - 3 = 0$
(C) $2x + y - 3 = 0$
(D) $2x - y + 3 = 0$
2. What is the domain of the function $f(x) = \sqrt{x^2 - 2x}$? 1
- (A) $x \geq 2$
(B) $x > 2$
(C) $x \leq 0$ or $x \geq 2$
(D) $x < 0$ or $x > 2$
3. For what values of k does the polynomial $P(x) = (k^3 - k)x^4 + (k^2 - k)x^3 + x^2$ have degree 3? 1
- (A) $k = -1$
(B) $k = -1$ or $k = 0$
(C) $k = -1$ or $k = 1$
(D) $k = -1, k = 0$ or $k = 1$
4. What is the remainder when $P(x) = x^2 - (k+1)x - (k+1)^2$ is divided by $(x+k)$? 1
- (A) $k^2 + k + 1$
(B) $k^2 - k - 1$
(C) $k^2 + k - 1$
(D) $k^2 - k + 1$
5. What is the value of $\cos^{-1}(\cos \frac{5\pi}{6}) - \sin^{-1}(\sin \frac{5\pi}{6})$? 1
- (A) 0
(B) $\frac{\pi}{3}$
(C) $-\frac{\pi}{2}$
(D) $\frac{2\pi}{3}$

Marks
16. Which of these is NOT an expression for $\tan 2x$?

- (A) $\frac{\sin 4x}{1 + \cos 4x}$
- (B) $\frac{1 - \cos 4x}{\sin 4x}$
- (C) $\frac{1}{1 - \tan x} - \frac{1}{1 + \tan x}$
- (D) $\frac{\tan x}{1 - \tan x} - \frac{\tan x}{1 + \tan x}$

7. A group consists of 3 pairs of brothers and sisters from 3 different families. In how many ways can the group be seated in a circle so that the members of the same family are sitting opposite each other?

1

- (A) 8
- (B) 12
- (C) 16
- (D) 20

8. Which of the following is obtained by expanding $(2 + \sqrt{3})^3$?**1**

- (A) $8 + 3\sqrt{3}$
- (B) $14 + 7\sqrt{3}$
- (C) $20 + 11\sqrt{3}$
- (D) $26 + 15\sqrt{3}$

47 Marks**Marks****Attempt Questions 9-25****Allow about 1 hour and 18 minutes for this section.**

Answer the questions in the spaces provided. Sufficient space is provided for typical responses.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided at the back of this booklet.

If you use this space, clearly indicate which question you are answering

Question 9 (1 mark)Solve the inequality $|2x - 1| \leq 3$.**1**

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Question 10 (3 marks)Find the coefficient of the term containing x^3 in the expansion of $\left(2x + \frac{3}{x}\right)^{11}$.**3**

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Question 11 (2 marks)Show that $\frac{\cos^2 x - \cos 2x}{\sin^2 x + \cos 2x} = \tan^2 x$.**2**

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Question 12 (3 marks)Solve the inequality $\frac{x^2-2}{x} < 1$.

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Question 13 (3 marks)Consider the polynomial $P(x) = x^3 - px^2 + qx - pq$, where $q > 0$.(a) Show that $(x - p)$ is a factor of $P(x)$.

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(b) Show that the equation $P(x) = 0$ has only one real root.

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Question 14 (3 marks)

Marks

Solve the equation $\sin 2x = \sin x$ for $0 \leq x \leq 2\pi$.

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Question 15 (3 marks)

Find the number of ways that a team of 4 can be chosen from 5 boys and 6 girls

(a) with equal numbers of boys and girls.

1

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(b) with a majority of girls.

2

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Question 16 (3 marks)

The equation $x^3 + 6x^2 + 4x + 2 = 0$ has roots α , β and γ .

(a) Find the value of $\frac{\alpha\beta}{2} + \frac{\beta\gamma}{2} + \frac{\gamma\alpha}{2}$.

1

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(b) Find the value of $\frac{2}{\alpha\beta} + \frac{2}{\beta\gamma} + \frac{2}{\gamma\alpha}$.

2

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Question 17 (3 marks)

- (a) Find the domain and range of the function $f(x) = 2\cos^{-1}(x-1)$. 2

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- (b) Sketch the graph of the function $f(x) = 2\cos^{-1}(x-1)$ showing the intercepts on the axes and the coordinates of the endpoints. 1

Question 18 (3 marks)

- (a) Write down the binomial expansion of $(1+x)^n$. 1

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- (b) Hence show that ${}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n = 2^n - 1$. 2

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Student name / number _____

Marks

Question 19 (2 marks)

Find the multiplicity of the root 1 of the equation $P(x) = 0$ where

2

$$P(x) = x^4 - 5x^3 + 9x^2 - 7x + 2.$$

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Question 20 (2 marks)

Sketch the graph of the function $f(x) = \tan^{-1} x$ showing the intercepts on the axes and the equations of the asymptotes.

2

Question 21 (4 marks)

- (a) Find in fully expanded form the Cartesian equation of the circle with parametric equations 2

$$x = 4 + 5\cos\theta \quad \text{and} \quad y = 3 + 5\sin\theta.$$

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- (b) Sketch the graph of this circle showing the intercepts on the axes. 2

Question 23 (2 marks)

Find the number of ways in which the letters of the word HEXAGON can be arranged in a straight line

(a) so that the vowels are all next to each other.

1

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(b) so that no two consonants are next to each other

1

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Question 24 (2 marks)

If $\cos^{-1} x - \sin^{-1} x = \frac{\pi}{4}$, find the exact value of $\cos^{-1} x$.

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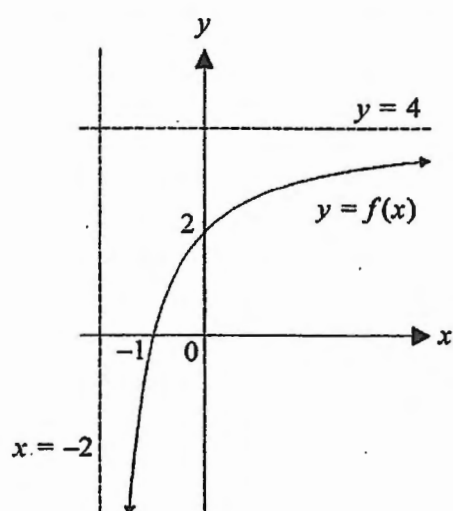
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Question 25 (4 marks)

The diagram shows the graph of the curve $y = f(x)$ where $f(x) = \frac{4x+4}{x+2}$, $x > -2$.

On separate diagrams sketch the graphs of the following curves, showing clearly the intercepts on the axes and the equations of any asymptotes.



(a) $y = |f(x)|$

1

(b) $y^2 = f(x)$

1

(c) $y = \frac{1}{f(x)}$

2

Student name / number _____

Marks

Section II Extra writing space.

If you use this space, clearly indicate which question you are answering.

[illegible]

Student name / number _____

Marks

[illegible]



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3	X			
4		X		
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6				X
7	X			
8				X

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3. For what values of k does the polynomial $P(x) = (k^3 - k)x^4 + (k^2 - k)x^3 + x^2$ have degree 3? 1
 - (A) $k = -1$
 - (B) $k = -1$ or $k = 0$
 - (C) $k = -1$ or $k = 1$
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4. What is the remainder when $P(x) = x^2 - (k+1)x - (k+1)^2$ is divided by $(x+k)$? 1
 - (A) $k^2 + k + 1$
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5. What is the value of $\cos^{-1}\left(\cos \frac{5\pi}{6}\right) - \sin^{-1}\left(\sin \frac{5\pi}{6}\right)$? 1
 - (A) 0
 - (B) $\frac{\pi}{3}$
 - (C) $-\frac{\pi}{2}$
 - (D) $\frac{2\pi}{3}$

Marks
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(D) $\frac{\tan x}{1 - \tan x} - \frac{\tan x}{1 + \tan x}$

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Question 9 (1 mark)Solve the inequality $|2x - 1| \leq 3$.**1**

$$-3 \leq 2x - 1 \leq 3$$

$$-2 \leq 2x \leq 4$$

$$-1 \leq x \leq 2$$

Question 10 (2 marks)Find the coefficient of the term containing x^3 in the expansion of $(2x + \frac{3}{x})^{11}$.**3**

$$(2x + \frac{3}{x})^{11} = {}^{11}C_0 (2x)^0 (\frac{3}{x})^{11} + {}^{11}C_1 (2x)^1 (\frac{3}{x})^{10} + {}^{11}C_2 (2x)^2 (\frac{3}{x})^9 + \dots$$

$$\begin{aligned} \text{Term containing } x^3 \text{ is } & {}^{11}C_7 (2x)^7 (\frac{3}{x})^4 \\ &= {}^{11}C_7 \times 2^7 x^7 \times 3^4 x^{-4} \\ &= {}^{11}C_7 \times 2^7 \times 3^4 \times x^3 \\ &= 3421440 x^3 \end{aligned}$$

 $\therefore$ coefficient is 3421440**Question 11 (2 marks)**Show that $\frac{\cos^2 x - \cos 2x}{\sin^2 x + \cos 2x} = \tan^2 x$.**2**

$$\begin{aligned} \frac{\cos^2 x - \cos 2x}{\sin^2 x + \cos 2x} &= \frac{\cos^2 x - (\cos^2 x - \sin^2 x)}{\sin^2 x + (\cos^2 x - \sin^2 x)} \\ &= \frac{\sin^2 x}{\cos^2 x} \\ &= \tan^2 x \end{aligned}$$

Question 12 (3 marks)Solve the inequality $\frac{x^2-2}{x} < 1$.

3

$$\frac{x^2-2}{x} < 1$$

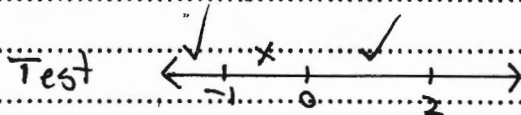
$$x \neq 0$$

$$x(x^2-2) < x^2$$

$$x(x^2-2)-x^2 < 0$$

$$x(x^2-x-2) < 0$$

$$x(x+1)(x-2) < 0$$



$$\therefore x < -1 \text{ or } 0 < x < 2$$

Question 13 (3 marks)Consider the polynomial $P(x) = x^3 - px^2 + qx - pq$, where $q > 0$.(a) Show that $(x-p)$ is a factor of $P(x)$.

1

$$\begin{aligned} P(p) &= p^3 - p \cdot p^2 + qp - pq \\ &= 0 \end{aligned}$$

$\therefore x-p$ is a factor

(b) Show that the equation $P(x) = 0$ has only one real root.

2

$$\begin{aligned} P(x) &= x^3 - px^2 + qx - pq \\ &= x^2(x-p) + q(x-p) \\ &= (x^2+q)(x-p) \end{aligned}$$

now $x^2+q=0$ has no real roots

$\therefore P(x)=0$ has only one real root p

Question 14 (3 marks)

Solve the equation $\sin 2x = \sin x$ for $0 \leq x \leq 2\pi$.

3

$$\sin 2x = \sin x \quad 0 \leq x \leq 2\pi$$

$$2\sin x \cos x = \sin x$$

$$\sin x(2\cos x - 1) = 0$$

$$\sin x = 0 \text{ or } \cos x = \frac{1}{2}$$

$$x = 0, \pi, 2\pi \quad x = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\therefore x = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi$$

Question 15 (3 marks)

Find the number of ways that a team of 4 can be chosen from 5 boys and 6 girls

(a) with equal numbers of boys and girls.

1

$${}^5C_2 \times {}^6C_2 = 150$$

(b) with a majority of girls.

2

$${}^6C_4 \times {}^5C_0 + {}^6C_3 \times {}^5C_1 = 115$$

Question 16 (3 marks)

The equation $x^3 + 6x^2 + 4x + 2 = 0$ has roots α , β and γ .(a) Find the value of $\frac{\alpha\beta}{2} + \frac{\beta\gamma}{2} + \frac{\gamma\alpha}{2}$.

1

$$\alpha\beta + \beta\gamma + \gamma\alpha = 4$$

$$\therefore \frac{\alpha\beta}{2} + \frac{\beta\gamma}{2} + \frac{\gamma\alpha}{2} = 2$$

(b) Find the value of $\frac{2}{\alpha\beta} + \frac{2}{\beta\gamma} + \frac{2}{\gamma\alpha}$.

2

$$\frac{2}{\alpha\beta} + \frac{2}{\beta\gamma} + \frac{2}{\gamma\alpha} = \frac{2(\gamma + \alpha + \beta)}{\alpha\beta\gamma}$$

$$= \frac{2 \times (-6)}{-2}$$

$$= 6$$

Marks

Question 17 (3 marks)

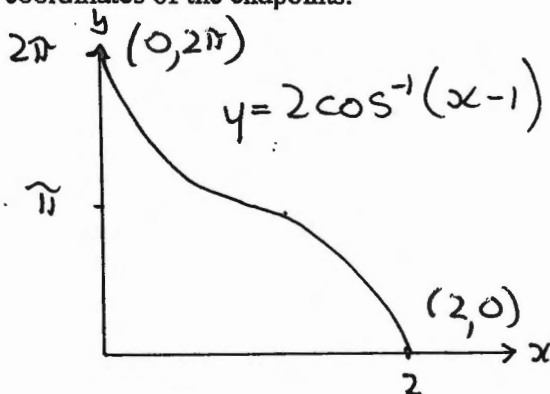
(a) Find the domain and range of the function $f(x) = 2\cos^{-1}(x-1)$.

2

Domain: $-1 \leq x-1 \leq 1$ Range: $0 \leq \cos^{-1}(x-1) \leq \pi$
 $\therefore 0 \leq x \leq 2$ $\therefore 0 \leq f(x) \leq 2\pi$

(b) Sketch the graph of the function $f(x) = 2\cos^{-1}(x-1)$ showing the intercepts on the axes and the coordinates of the endpoints.

1



Question 18 (3 marks)

(a) Write down the binomial expansion of $(1+x)^n$.

1

$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + \dots + {}^nC_nx^n$

(b) Hence show that ${}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n = 2^n - 1$.

2

substituting $x=1$, $2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n$

now ${}^nC_0 = 1 \therefore 2^n - 1 = {}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n$

Question 19 (2 marks)Find the multiplicity of the root 1 of the equation $P(x)=0$ where

2

$$P(x) = x^4 - 5x^3 + 9x^2 - 7x + 2$$

$$P(x) = x^4 - 5x^3 + 9x^2 - 7x + 2 \quad P(1) = 0$$

$$P'(x) = 4x^3 - 15x^2 + 18x - 7 \quad P'(1) = 0$$

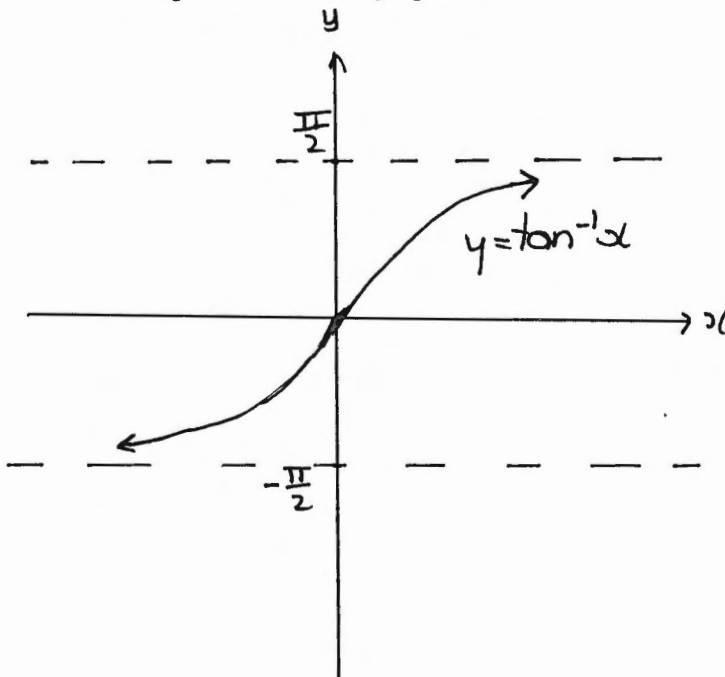
$$P''(x) = 12x^2 - 30x + 18 \quad P''(1) = 0$$

$$P'''(x) = 24x - 30 \quad P'''(1) \neq 0$$

$\therefore 1$ is a root of multiplicity of 3

Question 20 (2 marks)Sketch the graph of the function $f(x) = \tan^{-1} x$ showing the intercepts on the axes and the equations of the asymptotes.

2



Question 21 (4 marks)

(a) Find in fully expanded form the Cartesian equation of the circle with parametric equations

2

$$x = 4 + 5\cos\theta \text{ and } y = 3 + 5\sin\theta.$$

$$x = 4 + 5\cos\theta \quad y = 3 + 5\sin\theta$$

$$\cos\theta = \frac{x-4}{5} \quad \sin\theta = \frac{y-3}{5}$$

now

$$\cos^2\theta + \sin^2\theta = 1$$

$$\left(\frac{x-4}{5}\right)^2 + \left(\frac{y-3}{5}\right)^2 = 1$$

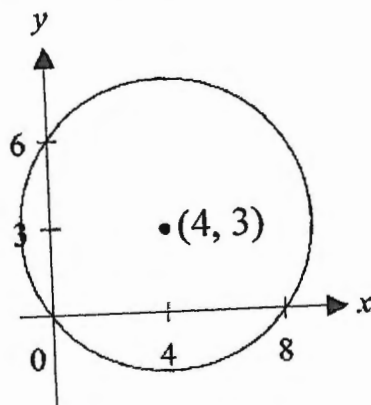
$$(x-4)^2 + (y-3)^2 = 25$$

$$x^2 - 8x + 4^2 + y^2 - 6y + 3^2 = 25$$

$$x^2 - 8x + 16 + y^2 - 6y + 9 = 25$$

(b) Sketch the graph of this circle showing the intercepts on the axes.

2



Question 22 (4 marks)

Consider the function $f(x) = x^2 - 2$, $0 \leq x \leq 4$.

- (a) Find the inverse function $f^{-1}(x)$ and state its domain.

2

$$y = x^2 - 2, \quad 0 \leq x \leq 4$$

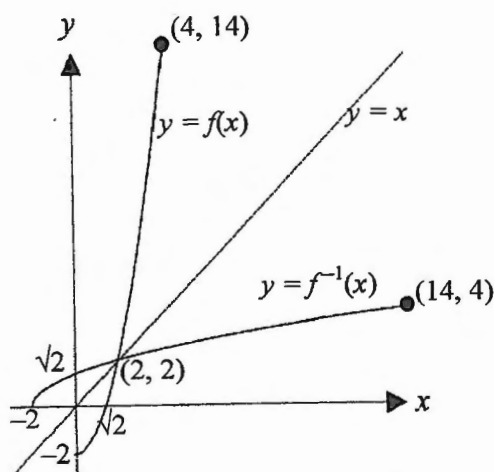
$$x^2 = y + 2$$

$$x = \sqrt{y + 2}, \quad -2 \leq y \leq 14$$

$$f^{-1}(x) = \sqrt{x + 2}, \quad -2 \leq x \leq 14$$

- (b) On the same diagram sketch the graphs of the function $f(x)$ and the inverse function $f^{-1}(x)$ showing the coordinates of the endpoints and the intercepts on the axes.

2



Question 23 (2 marks)

Find the number of ways in which the letters of the word HEXAGON can be arranged in a straight line

(a) so that the vowels are all next to each other. 1

(EAO) = vowels H X G N can be arranged in $5!$ ways.
 (EAO) can be arranged in $3!$ ways.
 $5! \times 3! = 720$ ways

(b) so that no two consonants are next to each other 1

V = vowels C = consonants
 can only be arranged as C V C V C V C V C
 C arrangements = $4!$ C arrangements = $3!$
 $4! \times 3! = 144$ ways

Question 24 (2 marks)

If $\cos^{-1} x - \sin^{-1} x = \frac{\pi}{4}$, find the exact value of $\cos^{-1} x$. 2

$$\cos^{-1} x - \sin^{-1} x = \frac{\pi}{4} \quad (1)$$

now $\cos^{-1} x + \sin^{-1} x = \frac{\pi}{2} \quad (2)$

(1) + (2)

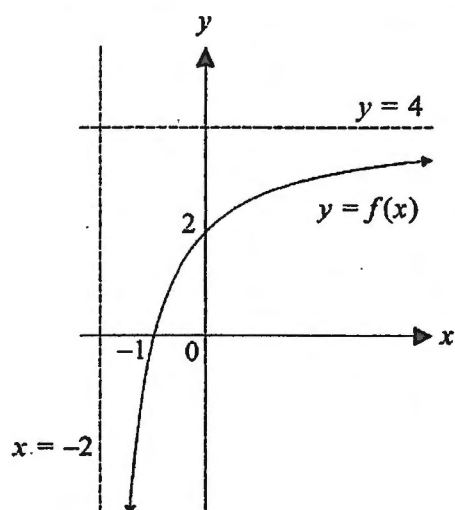
$$2\cos^{-1} x = \frac{3\pi}{4}$$

$$\cos^{-1} x = \frac{3\pi}{8}$$

Question 25 (4 marks)

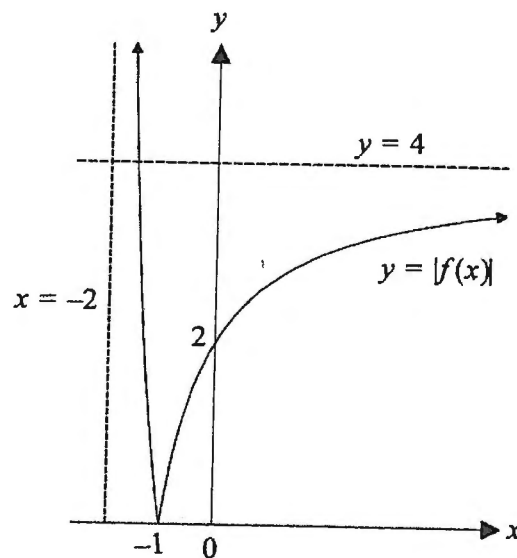
The diagram shows the graph of the curve $y = f(x)$ where $f(x) = \frac{4x+4}{x+2}$, $x > -2$.

On separate diagrams sketch the graphs of the following curves, showing clearly the intercepts on the axes and the equations of any asymptotes.



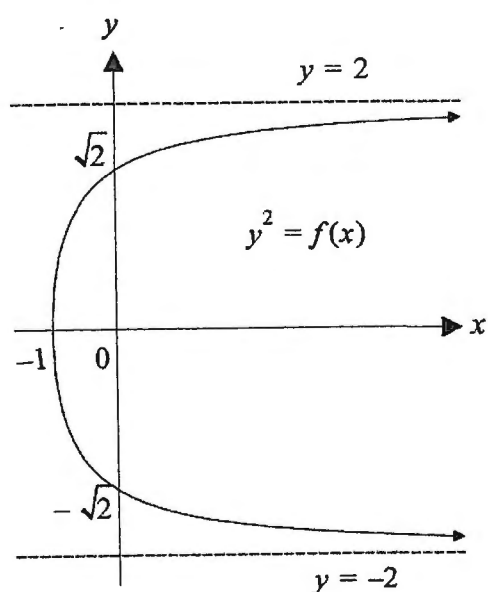
(a) $y = |f(x)|$

1



(b) $y^2 = f(x)$

1



(c) $y = \frac{1}{f(x)}$

2

